Supplementary Information — Designing for Emergence

SI 1. The rTCA engine

Network. The reductive tricarboxylic-acid (rTCA) cycle is modelled as an eight-component autocatalytic loop. Each intermediate x_i ($i = 1, \ldots, 8$) is regenerated over a full turn, and a completed turn fixes two molecules of CO_2 into the backbone, giving the network its autotrophic, self-reproducing character. The full stoichiometry and directed edge set (the row-normalized adjacency W used for structural measures) are built by `rTCA_network.build_W_rownorm`; node count $N = 8$.

Biochemical basis. The eight cycle intermediates, in the reductive direction, are oxaloacetate (OAA), malate (Mal), fumarate (Fum), succinate (Succ), succinyl-CoA (SuccoA), α-ketoglutarate (aKG), isocitrate (Icit), and citrate (Cit). A completed turn regenerates OAA while fixing two molecules of CO_2 (Table S1); acetyl-CoA (AcCoA) is supplied as the chemostat feed and constitutes the autotrophic boundary. Four steps are modelled with cooperative Hill kinetics ($n > 1$)—fumarate reductase (FRD), α-ketoglutarate:ferredoxin oxidoreductase (KGOR, the reductive counterpart of α-KGDH), isocitrate dehydrogenase (IDH), and ATP-citrate lyase (ACL)—reflecting their known allosteric regulation; the remaining four (malate dehydrogenase, fumarase, succinyl-CoA synthetase, aconitase) use Michaelis–Menten kinetics. The directed edge set and intermediate order follow KEGG map00720 (reductive TCA cycle); prebiotic, non-enzymatic variants of the same chemistry are reviewed in Muchowska et al. (2019).

Table S1. The rTCA engine: reductive-direction steps and kinetic regulation.

step	substrate → product	enzyme	kinetics
1	oxaloacetate → malate	malate dehydrogenase	Michaelis–Menten
2	malate → fumarate	fumarase	Michaelis–Menten
3	fumarate → succinate	fumarate reductase (FRD)	Hill, $n > 1$
4	succinate → succinyl-CoA	succinyl-CoA synthetase	Michaelis–Menten
5	succinyl-CoA → α-ketoglutarate	α-KGOR (reductive α-KGDH)	Hill, $n > 1$
6	α-ketoglutarate → isocitrate	isocitrate dehydrogenase (IDH)	Hill, $n > 1$
7	isocitrate → citrate	aconitase	Michaelis–Menten
8	citrate → oxaloacetate + acetyl-CoA	ATP-citrate lyase (ACL)	Hill, $n > 1$

Dynamics. Each component evolves under nonlinear Hill kinetics, $x_i = f_i(x; n) - \mu x_i$, where f_i is a saturating (Hill) production term driven by the cycle's predecessors with Hill coefficient n (the control parameter swept in the main text), and μx_i is a dilution/loss term. The nonlinearity n is the single knob that pushes the engine across its self-maintenance threshold. Hill functions, integration of the ODEs, and the maintenance (self-sustainability) functional are implemented in `emergence_net.py` (`hill`, `integrate`, `maintenance`); rate constants and reference concentrations are the constants in `emergence_model.py`.

Autotrophic boundary. A continuous feed of inorganic substrate (CO_2-derived) sustains the cycle against dilution; the feed strength is parameterized so that the *autotrophic* regime (feed from CO_2) can be contrasted with a *heterotrophic* regime (feed from organic precursor), the latter controlled by β_F in SI 6.

Self-maintenance threshold. T_{self} is located from the bistability of two ensembles—low- and high-initial-condition—integrated under the same n. Across the transition band, the low-init ensemble's time-to-collapse is bimodal: below threshold it collapses, above threshold it is recruited into the self-sustaining branch. The variance of the low-init outcome peaks at $T_{\mathrm{self}} \approx 3.75$ (Fig. 1A). A linearized control with identical connectivity and gains diverges instead of saturating, confirming the threshold is a nonlinear dynamical property (Fig. 1B). `run_rTCA.bistability_scan` / `find_T_self` implement the scan.

SI 2. Effective information and causal emergence

Transition-probability matrix (TPM). For each value of n, the running engine is discretized into B bins per component along its concentration range, giving B^8 micro-states. The dynamics are integrated for one step from each micro-state (or, equivalently, the deterministic flow is convolved with a small transition kernel), producing the transition-probability matrix $T_{ij} = P(s_j \mid s_i)$ over micro-states. `ce_measure.build_collective_tpm` constructs the collective TPM; `ce_measure.scan_ce` sweeps n.

Effective information. Under a uniform (maximum-entropy) intervention over micro-states, $P(s_i) = 1/N_s$, effective information is EI $= \frac{1}{N_s} \sum_i D_{\mathrm{KL}} \left[T(\cdot \mid s_i) \backslash \| T(\cdot)\right]$, where $T(\cdot) = \frac{1}{N_s} \sum_i T(\cdot \mid s_i)$ is the marginal effect. The do-uniform intervention guarantees EI depends only on the mechanism T, not on how often any input occurs. EI decomposes as determinism minus degeneracy: high determinism (each input maps reliably to one output) raises EI; high degeneracy (many inputs converge on the same output) lowers it. `ce_measure.effective_information` implements this; the form $Eff = EI/\log_2 N_s =$ determinism $-$ degeneracy is the standard Hoel normalization.

Coarse-graining and causal emergence. A coarse-graining maps micro-states to macro-states; the macro-TPM is the induced transition mechanism. Causal emergence is $\mathrm{EI}_{\mathrm{macro}} - \mathrm{EI}_{\mathrm{micro}} > 0$. The macro-level that maximizes EI is found by search over partitions (`ce_measure.locate_T_ce`); the located T_{CE} is the value of n at which emergence is maximal.

Calibration gate. Before use on rTCA, the implementation is checked against a test system with known EI. On the calibrated test: EI $= 0.811$ bit, $\Gamma = 1 + \sqrt{3}$ (the test's known reversibility), permuted $\Gamma = 4 = N$ (the degenerate permutation limit), giving CE $= 0.189$. The implementation reproduces these to numerical precision, confirming the TPM/EI pipeline is correct. `ce_measure._calibrate` runs this gate.

SI 3. Bootstrap of T_{self} and T_{CE}

To check that the coincidence of T_{self} and T_{CE} is not an artifact of a single fit, both thresholds are bootstrapped over resampled initial-condition ensembles. For each resample, T_{self} is recomputed from the low-init variance peak and T_{CE} from the EI maximum.

Result (Fig. S-Bootstrap; `data_bootstrap.npz`):

- $T_{\mathrm{CE}} = 3.50$, 95% bootstrap interval [3.33, 3.57].
- $T_{\mathrm{self}} = 3.40$, 95% bootstrap interval [3.20, 3.79].

The intervals overlap; the 0.10–0.25 gap between point estimates is within bootstrap noise. The disorder-to-order transition and the causal-emergence sharpening are statistically co-located.

SI 4. Null models and the three-property signature

Three-property signature. Rather than claiming a peak EI value, we require a network to satisfy three simultaneous properties to "have the signature":

1. *Reliable* — the peak is reproduced under bootstrap resampling of the ensembles used to build the TPM.
2. *Located* — the peak occurs at a defined threshold (a clear maximum in n), not as a uniform plateau or a random spike.
3. *Graded* — EI rises monotonically through the threshold rather than jumping as an isolated point.

A network passes "ALL-3" only if all three hold. The criterion is deliberately stricter than a scalar threshold, precisely so that structural noise cannot easily satisfy it.

Null models (each 30 independent draws; `data_null_degree.npz`, `R5/sim/null_degree_preserving.py`):

- **NULL-A (degree-preserving).** Rewire edges while preserving every node's in-degree and out-degree. This destroys regulatory specificity but keeps the connectivity distribution.
- **NULL-B (allosteric permutation).** Keep the rTCA topology but permute the regulatory signs (activation/inhibition) among edges. This keeps the graph and degrees but destroys the signed mechanism.
- **ER (baseline).** Erdős–Rényi random graphs matched in size and edge density.

Result (Fig. 3):

network	ALL-3 pass rate	median peak EI
rTCA	100% (30/30)	0.849
NULL-B (allosteric)	30% (9/30)	—
NULL-A (degree)	6.7% (2/30)	0.850
ER	≈ 0%	—

The single-EI trap. NULL-A's median peak EI (0.850) is essentially identical to rTCA's (0.849), yet only 2/30 NULL-A networks pass ALL-3. A criterion based on the scalar alone would misclassify NULL-A as reproducing the phenomenon; the located-and-graded structure does not. This is the empirical justification for the three-property signature.

SI 5. Reversibility index Γ/B

The reversibility index measures how irreversibly the macro-dynamics compress micro-state information. Given the singular values $\{\sigma_k\}$ of the effective dynamics operator (the SVD of the macro-level transition structure, computed in `ce_measure.gamma_alpha`), define $\Gamma = (\sum_k \sigma_k^p)^{1/p}, \Gamma/B =$ Γnormalized by bin count B, with p set so that Γ captures the effective rank (we report the normalized form Γ/B). $\Gamma/B \to$ high when the dynamics are concentrated on a low-dimensional, causally specific attractor; $\Gamma/B \to$ low when dynamics are diffuse.

Within rTCA (`data_rTCA_phase3.npz`, `gnorm_rTCA`): Γ/B rises from 0.267 below threshold to 0.355 above, tracking the EI and effective-rank jumps at $T_{\mathrm{CE}} \approx 3.50$. Effective rank rises 1 → 3 (with a brief 4 in the transition band).

SI 6. Feed-withdrawal (autotrophic-boundary) protocol

To test the autotrophic boundary, the heterotrophic feed fraction β_F is swept while holding the loop intact. The engine is run at each (n, β_F) and the peak EI over n is recorded (`data_rTCA_phase4_finefeed.npz`).

β_F	**0.004**	**0.005**	**0.006**	**0.007**	**0.008**	**0.009**	**0.010**	**0.012**	**0.014**	**0.02+**
peak EI	0.856	0.848	0.864	0.820	0.730	0.481	0.244	0.024	~0	~0

The signature survives in a narrow autotrophic window ($\beta_F \lesssim 0.009$) and collapses to zero by $\beta_F \gtrsim 0.012$–0.014. Removing the autotrophic boundary abolishes the transition even though the closed loop is intact (Fig. 4B).

SI 7. Multi-compartment integration

Two copies of the engine are coupled along two channels, and the integrated peak EI is compared to the sum of the isolated peaks (`data_rTCA_phase5.npz`; `phase5_integration.alpha_eff` / `gamma_eff`):

- **Base** (no coupling): peak EI 0.801.
- **Compartmentalization** (coupling across a compartment boundary): peak EI 0.967; threshold shifts lower, $T_{\mathrm{CE}} \approx 2.50$.
- **Energy coupling** (coupling via a shared energy carrier): peak EI 0.803; *no* threshold shift.
- **Both**: peak EI 0.967.

Superadditivity. The additive prediction for the integrated peak (sum of parts, adjusted) is 0.969; the observed (both-channel) peak is 0.967. Superadditivity $= 0.967 - 0.969 = -0.002$: integration is additive, not synergistic. Compartmentalization shifts the threshold and raises the peak but does so by linear superposition, not by creating new emergence (Fig. 5).

Graded sweep. Over integration strength $\theta \in [0,0.6]$, peak EI rises smoothly from 0.856 to 0.976, confirming the integration effect is graded rather than a discrete jump.

SI 8. Robustness

Bin-count sweep. T_{CE} and peak EI are recomputed across bin counts $B \in \{10,12,14,16\}$ (Fig. S-bsensitivity; `data_rTCA_phase4.npz` family). $T_{\mathrm{CE}} = 3.50$ is stable across B, and peak EI remains ≈ 0.84. The threshold is a dynamical property, not a binning artifact.

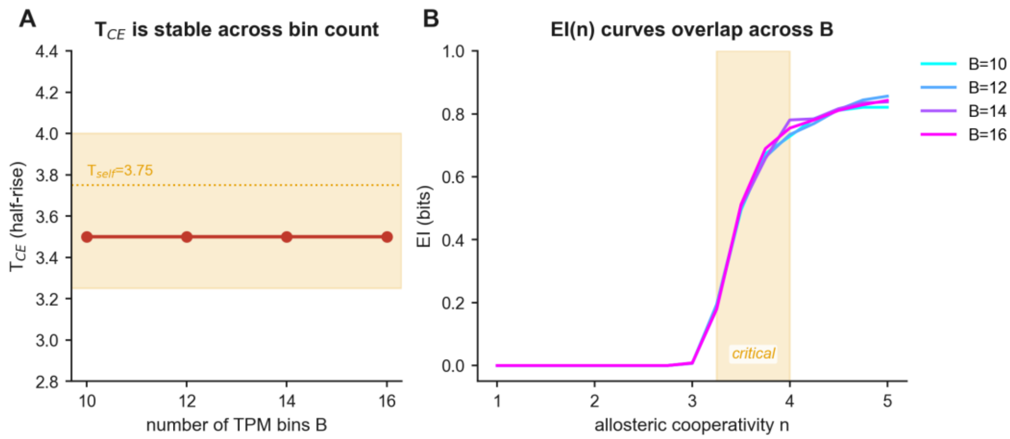

Figure S-bsensitivity. T_CE (A) and EI(n) curves (B) are stable across TPM bin count B.

Noise regime. The perturbations used here act as *observation* noise (they blur the read state without altering the flow), which tends to *inflate* EI; *dynamical* noise (added to the flow itself) tends to *deflate* EI. We report results in the observation-noise regime and flag this explicitly: a full dynamical-noise map is future work.

SI 9. Cross-network comparison: rTCA, Wood–Ljungdahl, syn3A

All three networks are run on the same engine machinery and measured identically (`data_rTCA_phase4ext.npz`).

network	topology	dom. eigenvalue	peak EI (bit)	T_{CE}	effective rank
rTCA	closed loop (autotrophic)	1.0	0.838	3.50	3
Wood–Ljungdahl (WLP)	acyclic (autotrophic)	0.0	0.071	— (no threshold)	2
syn3A core	acyclic (heterotrophic)	0.0	0.041	— (flat)	3

- **rTCA** is the only network with a closed loop (dominant eigenvalue 1.0, signal returns) *and* an autotrophic boundary; it is the only one with a sharp CE threshold.
- **WLP** is autotrophic but acyclic (signal flows through, eigenvalue 0.0); peak EI is 12× lower and shows no threshold.
- **syn3A** (core metabolism of the JCVI minimal cell: glycolysis + non-oxidative PPP + fermentation) is acyclic and heterotrophic; CE stays flat with no threshold (Fig. S-syn3A), consistent with the rule and extending the contrast from a designed engine to an evolved cell (with the caveat that syn3A differs from rTCA along two axes at once: acyclic and heterotrophic).

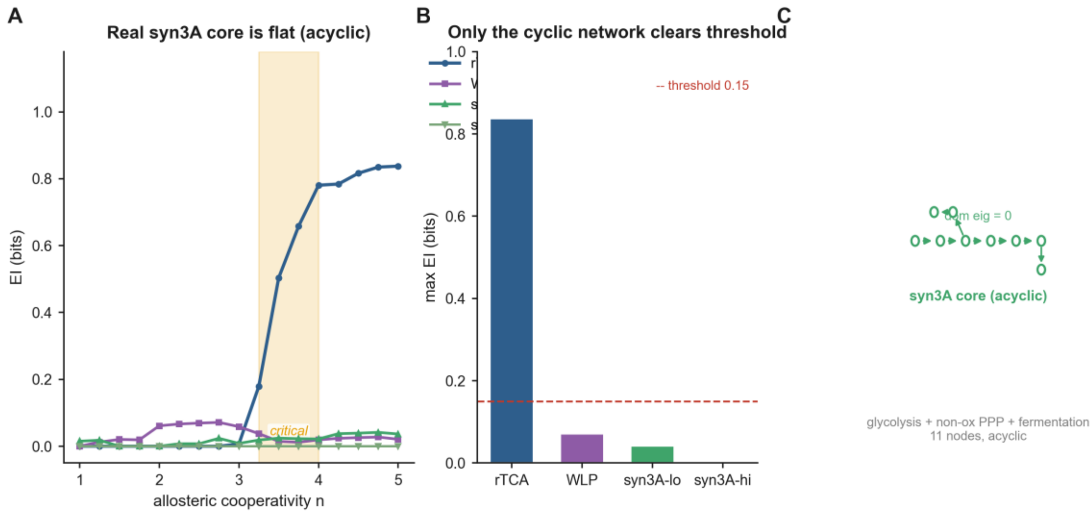

Figure S-syn3A. syn3A core cross-evaluation: EI versus n for four candidate networks (A), max-EI bars (B), and the syn3A acyclic topology (C).

syn3A topology is reconstructed from the essential-metabolism gene list of Breuer et al. (ref. 19); WLP topology from KEGG map00680 and the pathway review (ref. 22).

SI 10. Closest prior applications of causal emergence in biology

To support the claim that effective information has not previously been applied to the running dynamics of a self-sustaining metabolic cycle (Introduction; Materials and methods), we list the closest prior applications. The table is representative, not exhaustive, and is restricted to studies already cited in the main text.

study	system studied	what effective information / CE was applied to	why this is not "running self-sustaining metabolism"
Hoel et al., 2013; Hoel, 2017	toy / general systems	definition of EI; AND-gate toy calibration	foundational theory; no metabolic system
Klein and Hoel, 2020	biological / social / technological networks	CE of random-walk topology via subgraph grouping into macro-nodes	static connectivity, not dynamics
Rosas et al., 2020	multivariate data	partial-information decomposition (ΦID) of synergy	general data-driven decomposition; not specifically metabolic
Zaydman et al., 2022	bacterial proteomes (PPI)	SVD spectral emergence defining hierarchical interactions	static cross-proteome co-variation; not running metabolism
Pigozzi et al., 2025	gene-regulatory networks	integrative CE raised by associative conditioning	regulatory ODE dynamics; not autotrophically self-sustaining metabolism
Tkachenko and Maslov, 2024	prebiotic information-coding polymers	emergence of catalytic cleavage in a chemostat	information polymers, not metabolic chemistry

Across this set, effective information has been applied to network topology (Klein and Hoel, 2020), to gene regulation (Pigozzi et al., 2025), to static co-variation in proteomes (Zaydman et al., 2022), and to informationpolymer dynamics (Tkachenko and Maslov, 2024). To our knowledge it has not been applied to the running dynamics of a self-sustaining, autotrophic metabolic cycle—the object of the present work.

Data availability

All numerical results are computational; no new wet-laboratory data were generated. The primary data are saved NumPy `.npz` archives: the rTCA engine scans `data_rTCA_phase{1,2,3,4,4_finefeed,5,4ext}.npz`, the B-sensitivity sweep `data_rTCA_phase2_bsens.npz`, and the matched null `data_rTCA_phase3_null_matched.npz` (in `R3/sim/`); the bootstrap `data_bootstrap.npz` and the degree-preserving null `data_null_degree.npz` (in `R5/sim/`). Each file is documented in the SI section that analyses it. Figure source data correspond to these archives.

Code availability

The simulation and effective-information pipeline is the R3 codebase in `R3/sim/` (`emergence_model.py`, `emergence_net.py`, `ce_measure.py`, `topology.py`, `rTCA_network.py`, `WLP_network.py`, `syn3A_core_network.py`, `phase5_integration.py`, and the per-phase drivers `phase{1..5,4ext}_*.py`). The null-model analysis (`null_degree_preserving.py`), the bootstrap (`bootstrap_Tself_Tce.py`), and the Figure 3 regeneration script (`fig3_null_signature.py` in `R6/sim/`) build on this codebase, which is read-only and was not modified. The pipeline depends on NumPy, SciPy, and Matplotlib. Source files will be deposited in a public repository upon submission.

Key Resources Table

resource	source / identifier
Software: Python 3.10	https://www.python.org
Software: NumPy, SciPy, Matplotlib	https://numpy.org, https://scipy.org, https://matplotlib.org
rTCA network topology	KEGG map00720 (reductive TCA cycle)
Wood–Ljungdahl pathway topology	KEGG map00680; Ragsdale and Pierce, 2008
syn3A core metabolism	Breuer et al., 2019 (*eLife* 8:e36842)
rTCA simulation outputs (this paper)	`.npz` archives listed in *Data availability*
effective-information codebase (this paper)	`R3/sim/ce_measure.py` and related modules listed in *Code availability*